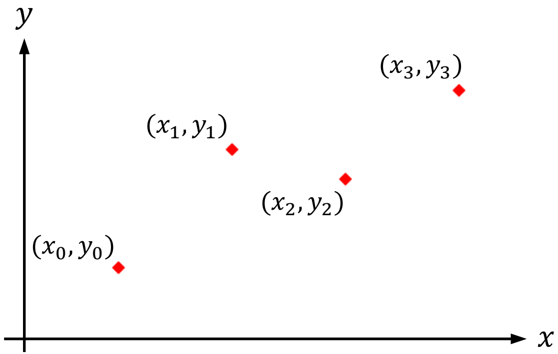
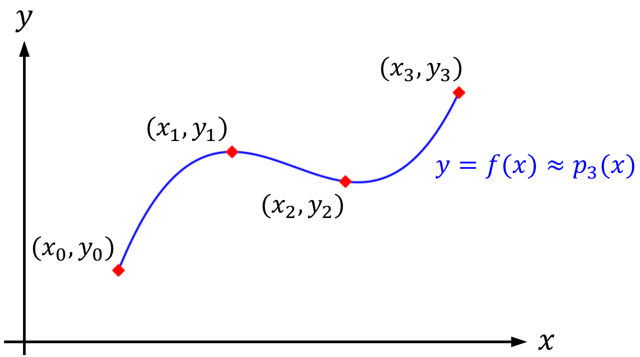
Approximation of Derivatives via Interpolation





Newton Interpolating Polynomials

$$p_n(x) = f[x_0] + f[x_0, x_1] \cdot (x - x_0) + f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1) + \cdots + f[x_0, x_1, \dots, x_n] \cdot (x - x_0) \cdot (x - x_1) \cdots (x - x_n); \quad x_0 \le x \le x_n$$

Lagrange Interpolating Polynomials

$$p_n(x) = L_0^{(n)} \cdot f(x_0) + L_1^{(n)} \cdot f(x_1) + \cdots + L_n^{(n)} \cdot f(x_n); \quad x_0 \le x \le x_n$$

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k^{th} Derivative

$$\frac{d^k}{dx^k} f(x) \approx \frac{d^k}{dx^k} p_n(x)$$

$\boxed{k = 1}$

$$\frac{d}{dx} f(x) \approx \frac{d}{dx} p_n(x)$$

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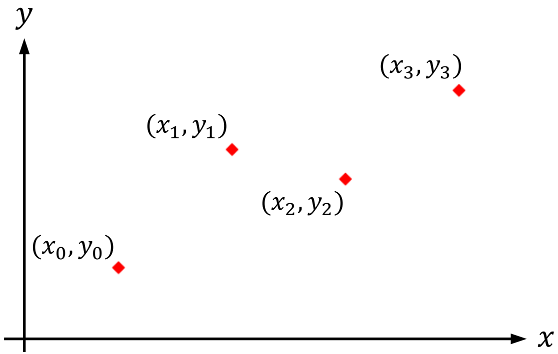

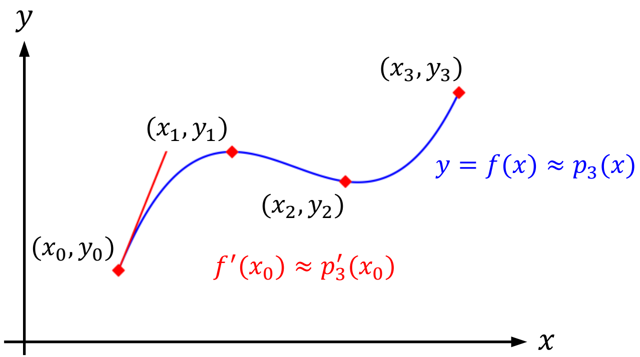

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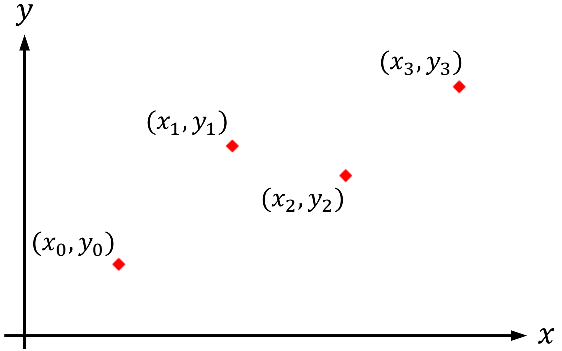

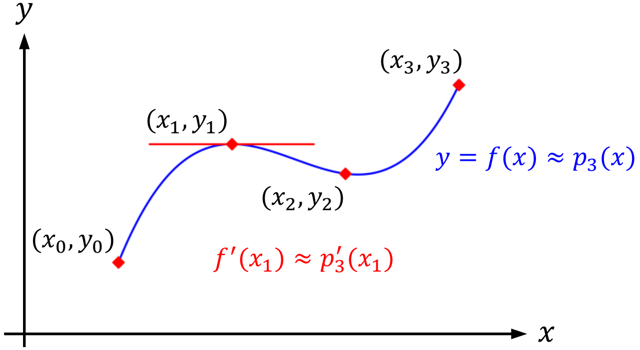

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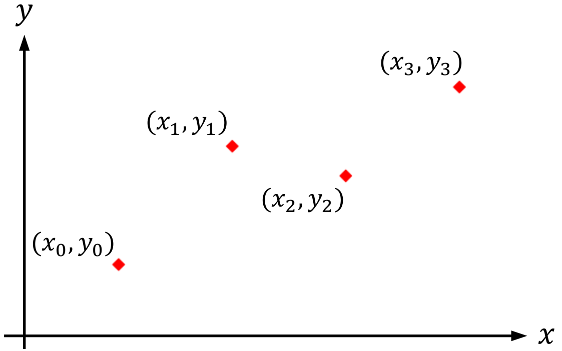

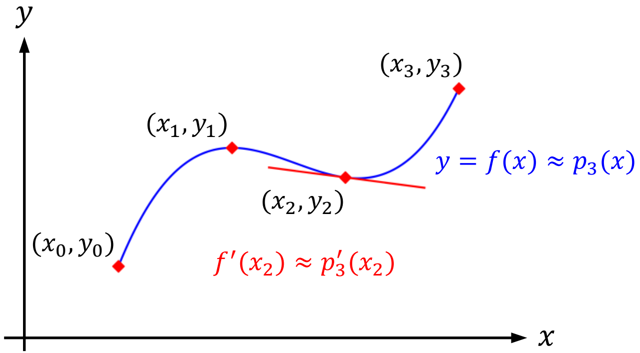

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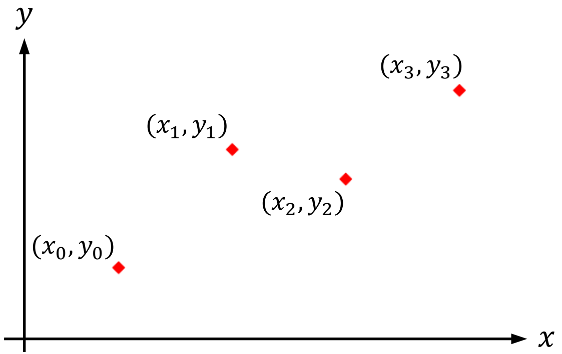

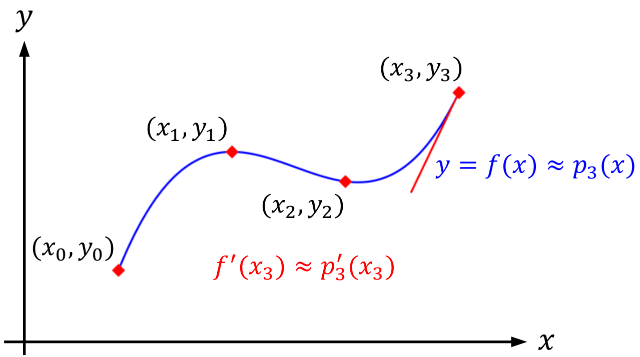

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